



Computer Networks Week # 12

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ACKNOWLEDGMENTS

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http://www.inetdaemon.com/tutorials/internet/ip/routing/interior_vs_exterior.shtml#sthash.dhDkYN0q.dpuf

Week -12 Course Plan

- Routing
- Forwarding techniques
- Concept of distance vector routing

Routing

- The process of moving data from one network to another is called **routing**
- Routing is unnecessary unless there are **multiple networks** on different address ranges
- If there are different combinations of IP addresses and subnet masks, So there are separate networks
- No need to run a routing protocol unless there are multiple networks
 - and even then, **manual static routes** are easier for small networks).

Routing

- Once we get beyond three routers, it's time to start thinking about **dynamic routing** protocols.
- If we are connecting our networks to the Internet,
 - we need to think about running **more than one kind of routing protocol**

Delivery of a Packet

- The network layer supervises the delivery of a packet
- The delivery of a packet to its final destination is accomplished by using **two different methods** of delivery
 - Direct delivery
 - Indirect delivery

Delivery of a Packet: Direct delivery

In a direct delivery

- Direct delivery occurs when the source and destination of the packet are located on the **same physical network**
- The sender can easily determine if the delivery is direct
 - It can extract the network address of the destination (using the mask) and
 - compare this address with the addresses of the networks to which it is connected
 - If a match is found, the delivery is direct

Delivery of a packet: Indirect Delivery

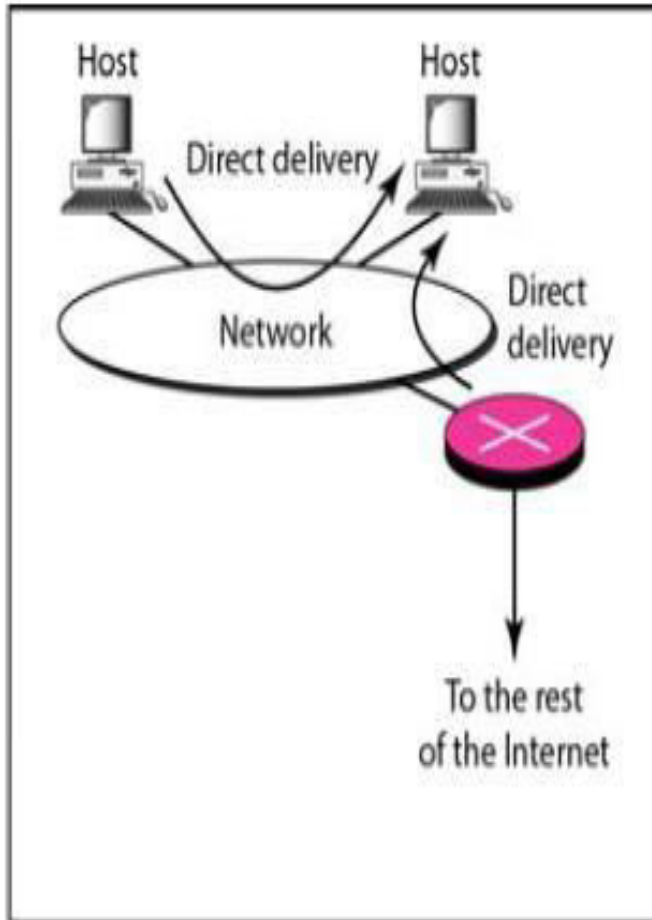
In Indirect delivery

- If the destination host is not on the same network as the deliverer, the packet is delivered indirectly
- The packet goes from router to router until
 - it reaches the one connected to the same physical network as its final destination.

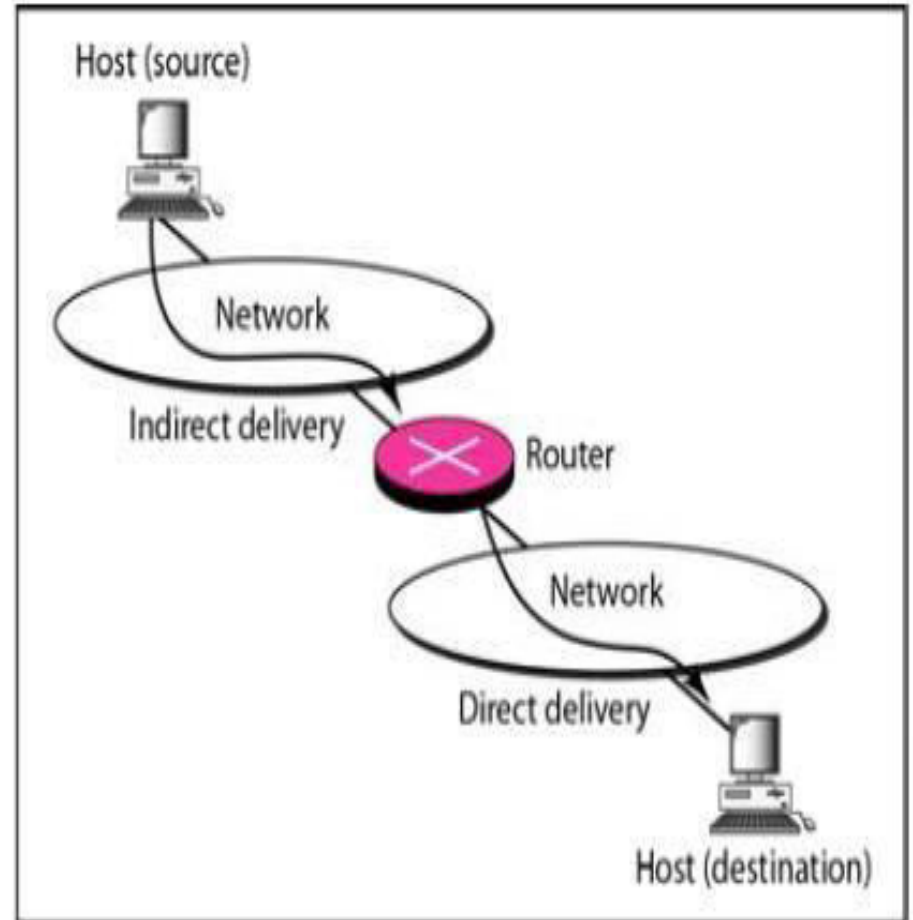
Note that

- a delivery always involves one direct delivery but zero or more indirect deliveries.
- the last delivery is always a direct delivery.

Direct and Indirect Delivery



a. Direct delivery



b. Indirect and direct delivery

Forwarding

Forwarding is the process of sending a packet on its way while

Routing is the process of deciding in which direction to send traffic

Forwarding is the process of getting through single interchange while

Routing is the process of planning trip from source to destination

For routing, we need **routing algorithm**

Forwarding Techniques

Next-hop Vs route method

Next-Hop: *A technique to reduce the contents of a routing table*

- The routing table holds only the address of the next hop instead of information about the complete route

Route method

- Information about the complete route are maintained

Forwarding Techniques

a. Routing tables based on route

Destination	Route
Host B	R1, R2, host B

Routing table
for host A

Destination	Route
Host B	R2, host B

Routing table
for R1

Destination	Route
Host B	Host B

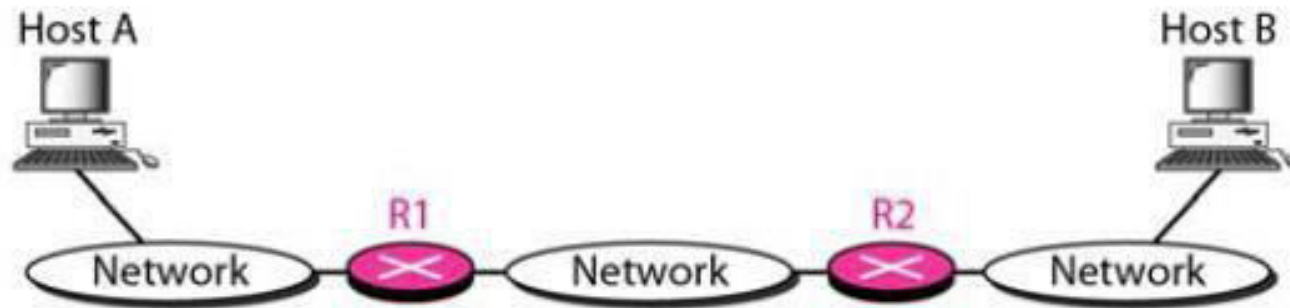
Routing table
for R2

b. Routing tables based on next hop

Destination	Next hop
Host B	R1

Destination	Next hop
Host B	R2

Destination	Next hop
Host B	---



Forwarding Techniques

Network-specific versus host-specific

Host-specific has an entry for every destination host connected to the same physical network

- The destination host address is given in routing table

Network-specific: A technique to reduce the routing table and simplify the searching process

- needs only one entry that defines the address of the destination network itself
- In other words, we treat all hosts connected to the same network as one single entity
- For example, if 1000 hosts are attached to the same network, only one entry exists in the routing table instead of 1000.

Forwarding Techniques

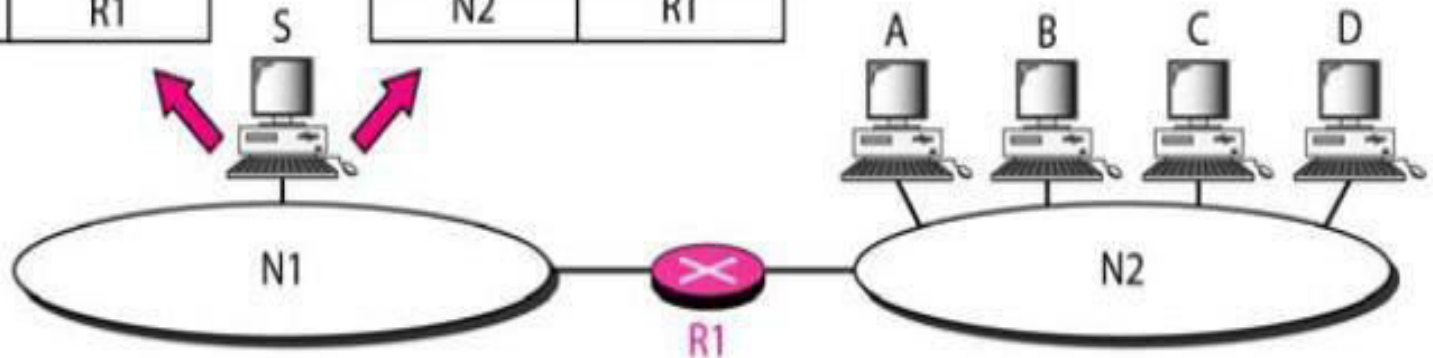
Network-specific versus host-specific

Routing table for host S based on host-specific method

Destination	Next hop
A	R1
B	R1
C	R1
D	R1

Routing table for host S based on network-specific method

Destination	Next hop
N2	R1



Forwarding Process

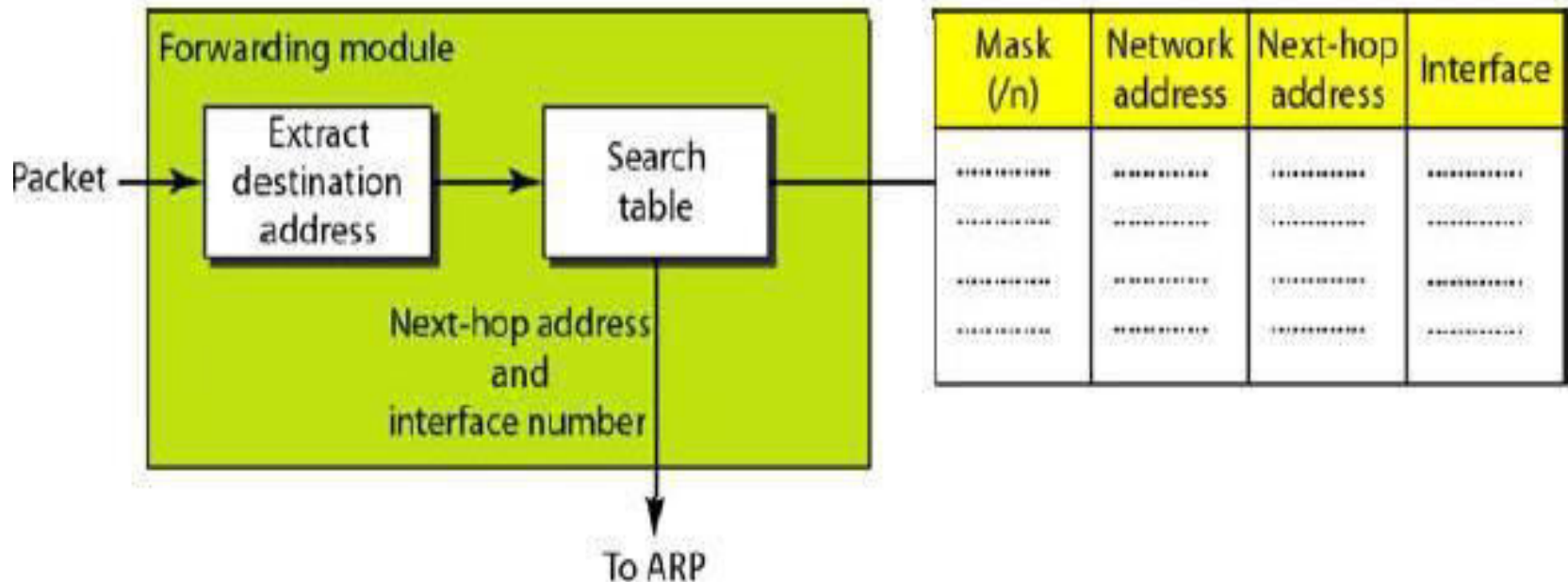
- Assume that hosts and routers use classless addressing because classful addressing can be treated as a special case of classless addressing.
- **In classless addressing**, the routing table needs to have one row of information for each block involved.

Problem: The table needs to be searched based on the network address (first address in the block)

- Unfortunately, the **destination address** in the packet gives no clue about the network address.

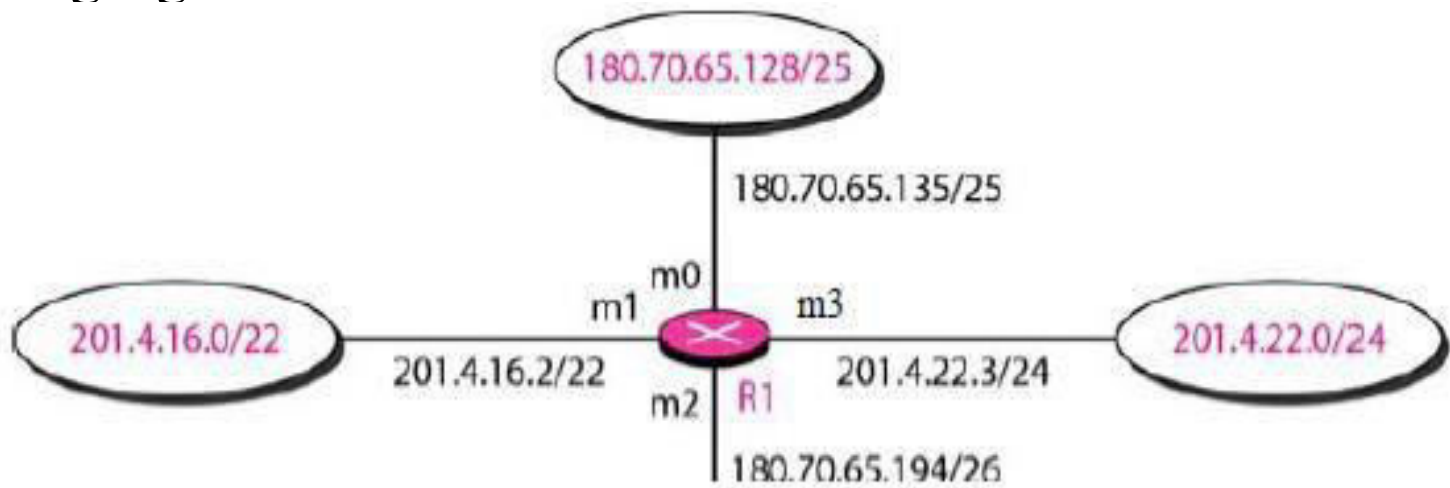
Forwarding Process

- To solve the problem, we need to include the mask (/n) in the table; we need to have an extra column that includes the mask for the corresponding block.
- A simple forwarding module for classless addressing is shown



Forwarding Process: Example

- Make a routing table for router R1, using the configuration in the following figure



<i>Mask</i>	<i>Network Address</i>	<i>Next Hop</i>	<i>Interface</i>
/26	180.70.65.192	—	m2
/25	180.70.65.128	—	m0
/24	201.4.22.0	—	m3
/22	201.4.16.0	...	m1
Any	Any	180.70.65.200	m2

Forwarding Process: Example

Q: Show the forwarding process if a packet arrives at R1 (Previous Figure) with the destination address 180.70.65.140.

Answer:

1: Applying the first mask (/26) does not match the corresponding network address.

2: The second mask (/25) is applied to the destination address. The result is 180.70.65.128, which matches the corresponding network address.

The next-hop address (the destination address of the packet in this case) and the interface number m0 are passed to ARP for further processing.

Forwarding Process: Example

Q: Show the forwarding process if a packet arrives at R1 (Previous Figure) with the destination address 201.4.22.35.

Answer:

The router performs the following steps:

1. The first mask (/26) is applied to the destination address. The result is 201.4.22.0, which does not match the corresponding network address (row 1).
2. The second mask (/25) is applied to the destination address. The result is 201.4.22.0, which does not match the corresponding network address (row 2).
3. **The third mask (/24) is applied to the destination address. The result is 201.4.22.0, which matches the corresponding network address.**
 - The destination address of the packet and the interface number m3 are passed to ARP.

Forwarding Process: Example

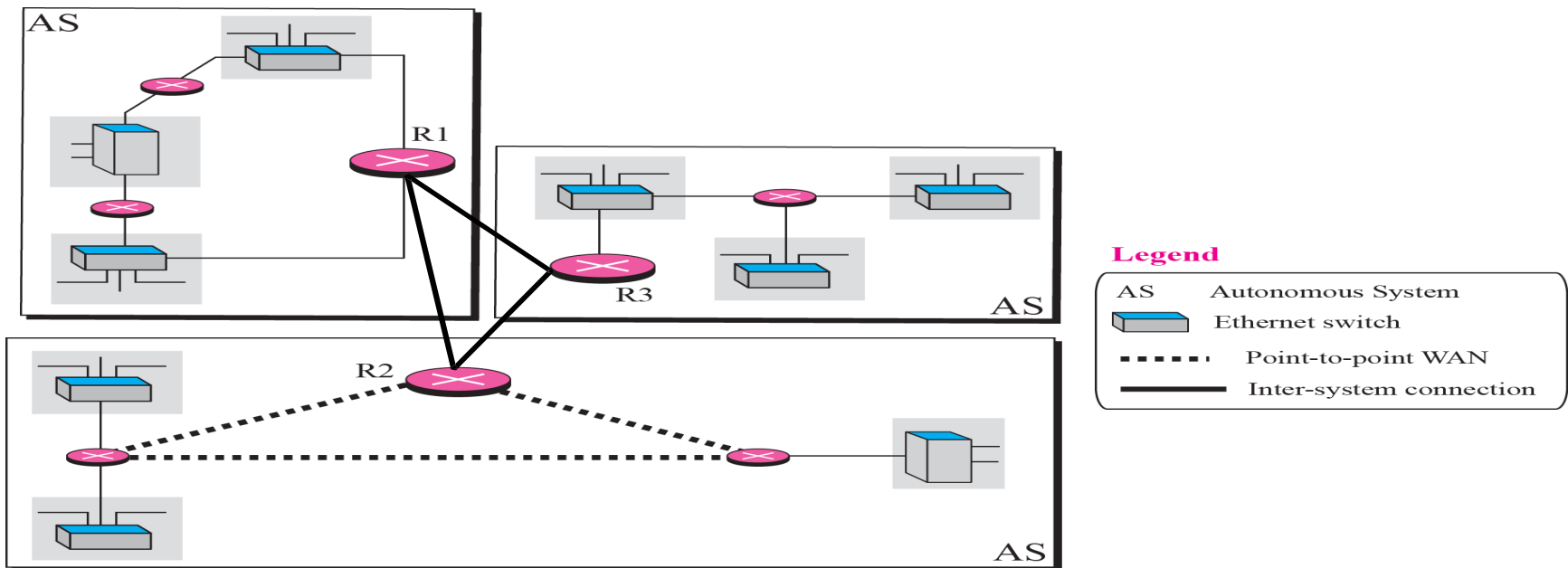
Q: Show the forwarding process if a packet arrives at R1 (Previous Figure) with the destination address 18.24.32.78.

Answer:

- **All masks are applied, one by one, to the destination address, but no matching network address is found.**
- When it reaches the end of the table, the module gives the next-hop address 180.70.65.200 and interface number m2 to ARP.
- This is probably an outgoing package that needs to be sent, via the default router, to someplace else in the Internet

Autonomous Systems

- Group of networks and routers under the authority of a single administration is called an **Autonomous System**
- Routing inside an autonomous system is called **intra-domain or Interior** routing.
- Routing between autonomous systems is called **inter-domain Exterior** routing

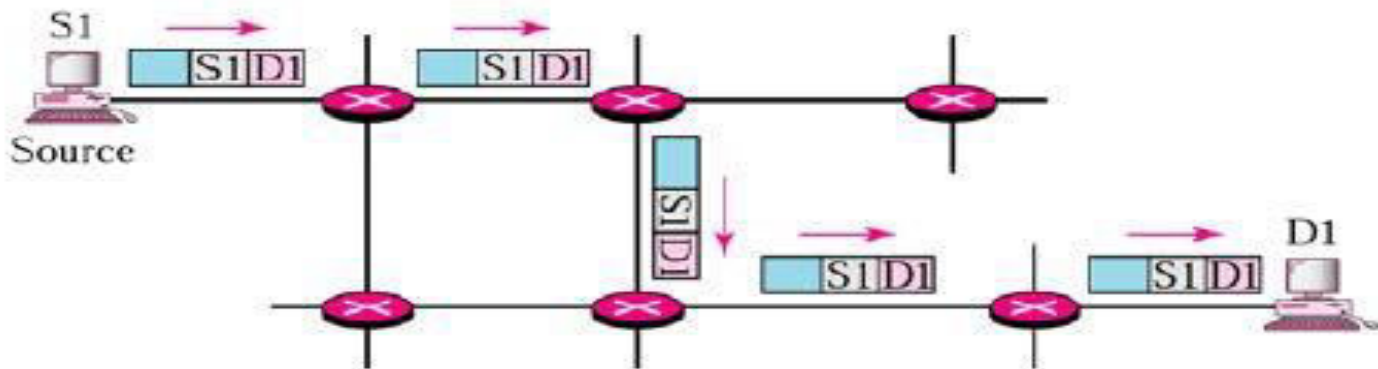


Unicast Routing Protocols

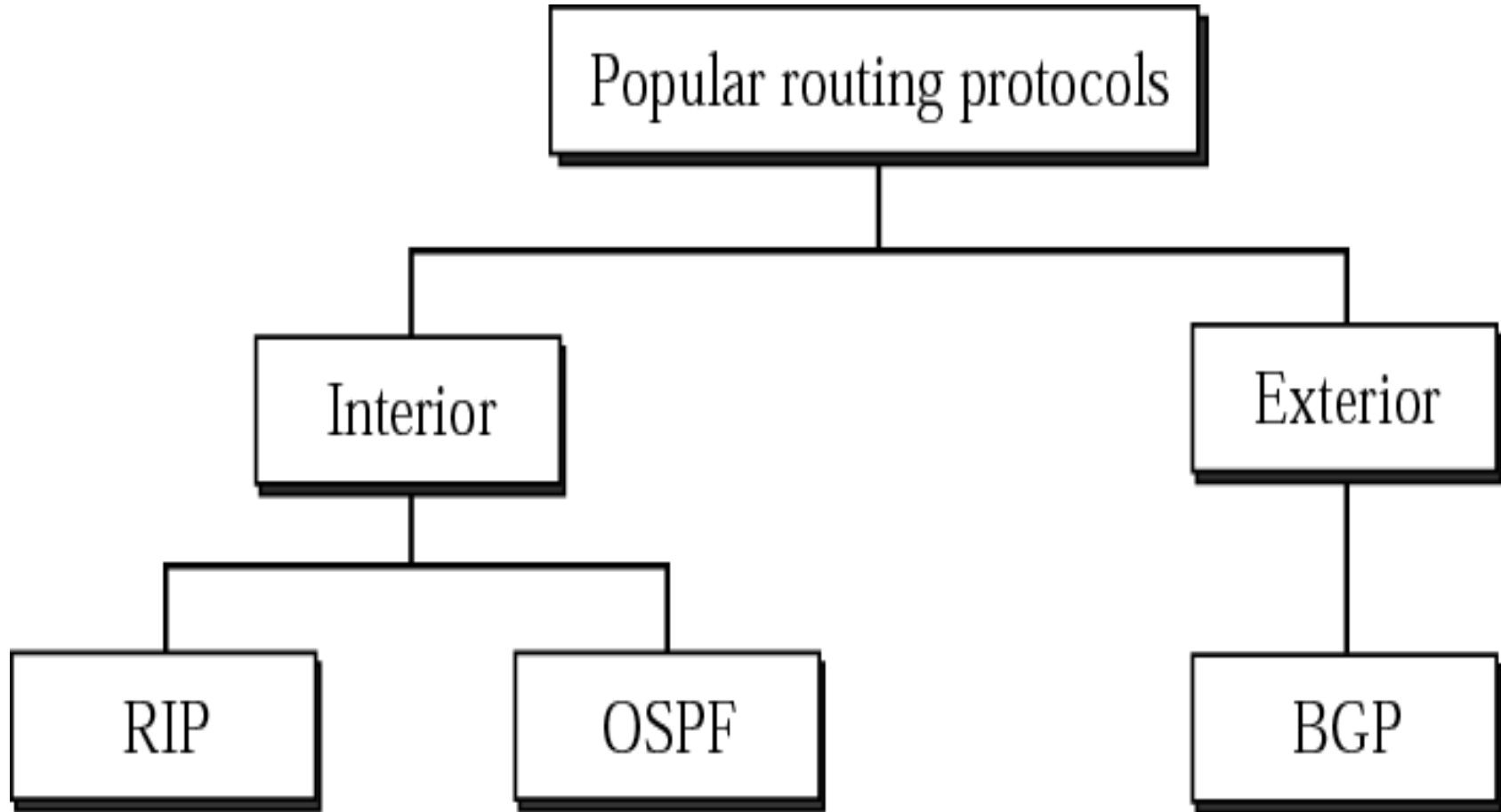
- **A routing table can be either static or dynamic.**
 - A static table is one with manual entries.
 - A dynamic table is one that is updated automatically when there is a change somewhere in the Internet.
- **A routing protocol** is a combination of rules and procedures that lets routers in the Internet inform each other of changes
- **Routing protocols are needed to create the routing tables dynamically**

Unicast Routing Protocols

- In unicast routing, the message is sent from one source to one destination (routed hop-by-hop).
- When a router receives a packet it forwards the packet through only one of its ports (the one belonging to the optimum path) as defined in routing table
- The packet may be discarded if the route couldn't be constructed



Popular Routing protocols



Interior Routing Protocols (IGP)

Routing inside an autonomous system

Categories

Distance vector protocols: router knows physically-connected neighbors, link costs to neighbors, iterative process of computation, exchange of partial information with neighbors

- Routing Information Protocol (**RIP**)

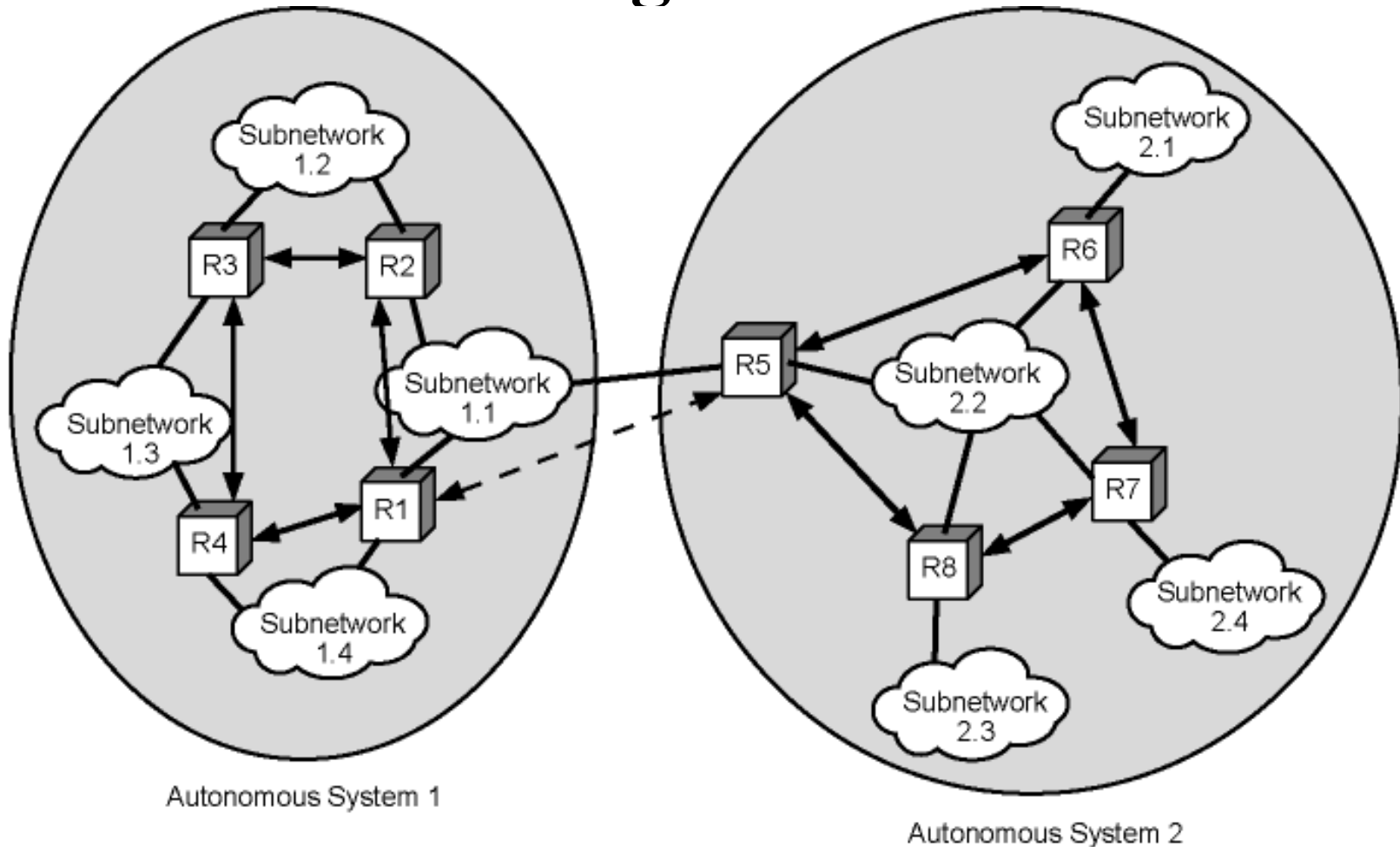
Link State Protocols: all routers have complete topology, link cost information

- Open Shortest Path First (**OSPF**)

Exterior routing Protocols (EGP)

- Routing between autonomous systems
- To get from place to place outside ones network(s), i.e. on the Internet, one must use an Exterior Gateway Protocol.
- EGPs handle routing outside an Autonomous System and get one from ones network, through ones Internet provider's network and onto any other network.
- **Examples of EGP**
 - Border Gateway Protocol (BGP) [A Path vector protocol]

Application of Exterior and Interior Routing Protocols



Metric of different Routing Protocols

Metric is the cost assigned for passing through a network

- The total metric of a particular route is equal to the sum of the metrics of networks that comprise the route.
- A router chooses the route with smallest metric.

RIP (Routing Information Protocol): the cost of passing each network is one hop count

- If a packet passes through 10 networks to reach the destination, the total cost is 10 hop counts.

Metric of different Routing Protocols

OSPF (Open Shortest Path First)

- Administrator can assign cost for passing a network based on type of service required.
- **OSPF** allows each router to have **more than one routing table** based on required type of **service**, such as Maximum throughput or minimum delay.

BGP (Border Gateway Protocol)

- Criterion is the policy, which is set by the administrator.

Distance Vector Routing

- Each node maintains a set of triples (Destination, Cost, NextHop)
- Node knows the cost to each neighbor
- Directly connected neighbors exchange updates
 - periodically (on the order of several seconds -30s-)
 - whenever table changes (called *triggered* update)
- Each update is a list of pairs:
 - (Destination, Cost)
- Update local table if a better route is received
 - smaller cost
 - came from next-hop
- Refresh existing routes; delete if timed out

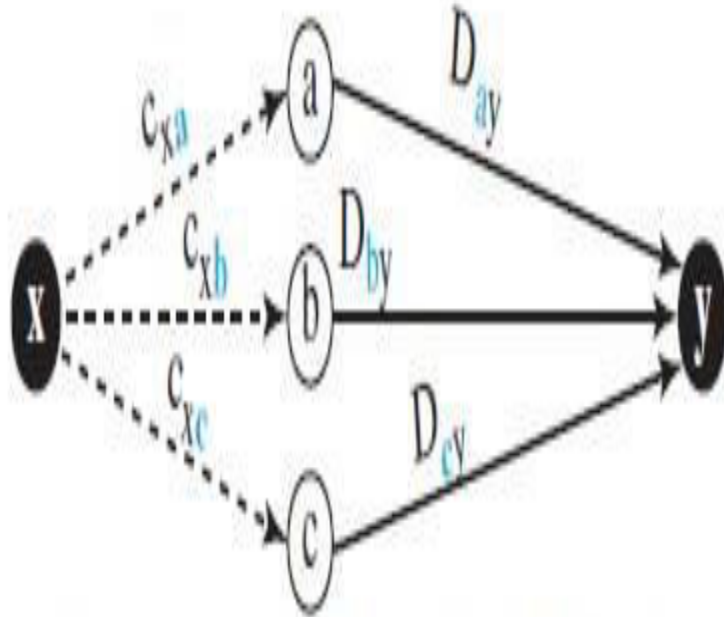
Bellman-Ford Equation

- Heart of distance vector routing
- Used to find the least cost (shortest distance) between a source node, x , and a destination node, y , through some intermediary nodes ($a, b, c, \dots$)
 - when the costs between the source and the intermediary nodes and the least costs between the intermediary nodes and the destination are given.

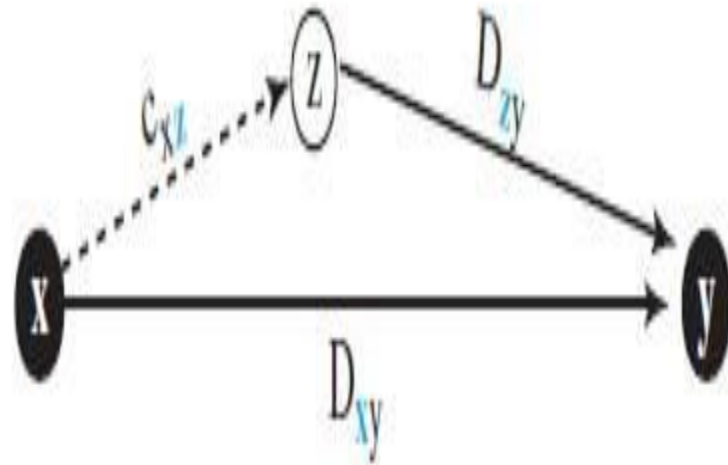
$$D_{xy} = \min\{(c_{xa} + D_{ay}), (c_{xb} + D_{by}), (c_{xc} + D_{cy}), \dots\}$$

- In distance-vector routing, normally update an existing least cost with a least cost through an intermediary node, such as z ,
 - if the latter is shorter.
- So the equation becomes simpler as $D_{xy} = \min\{D_{xy}, (c_{xz} + D_{zy})\}$

Bellman-Ford Equation

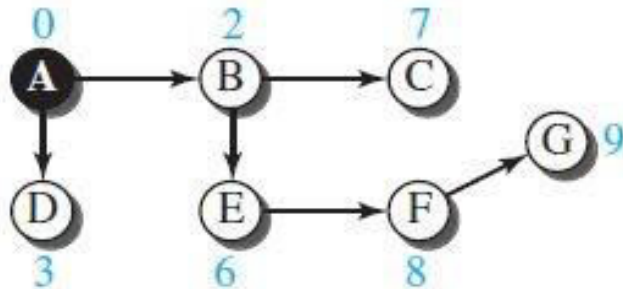


a. General case with three intermediate nodes



b. Updating a path with a new route

Distance Vector Corresponding to a Tree



a. Tree for node A

A	
A	0
B	2
C	7
D	3
E	6
F	8
G	9

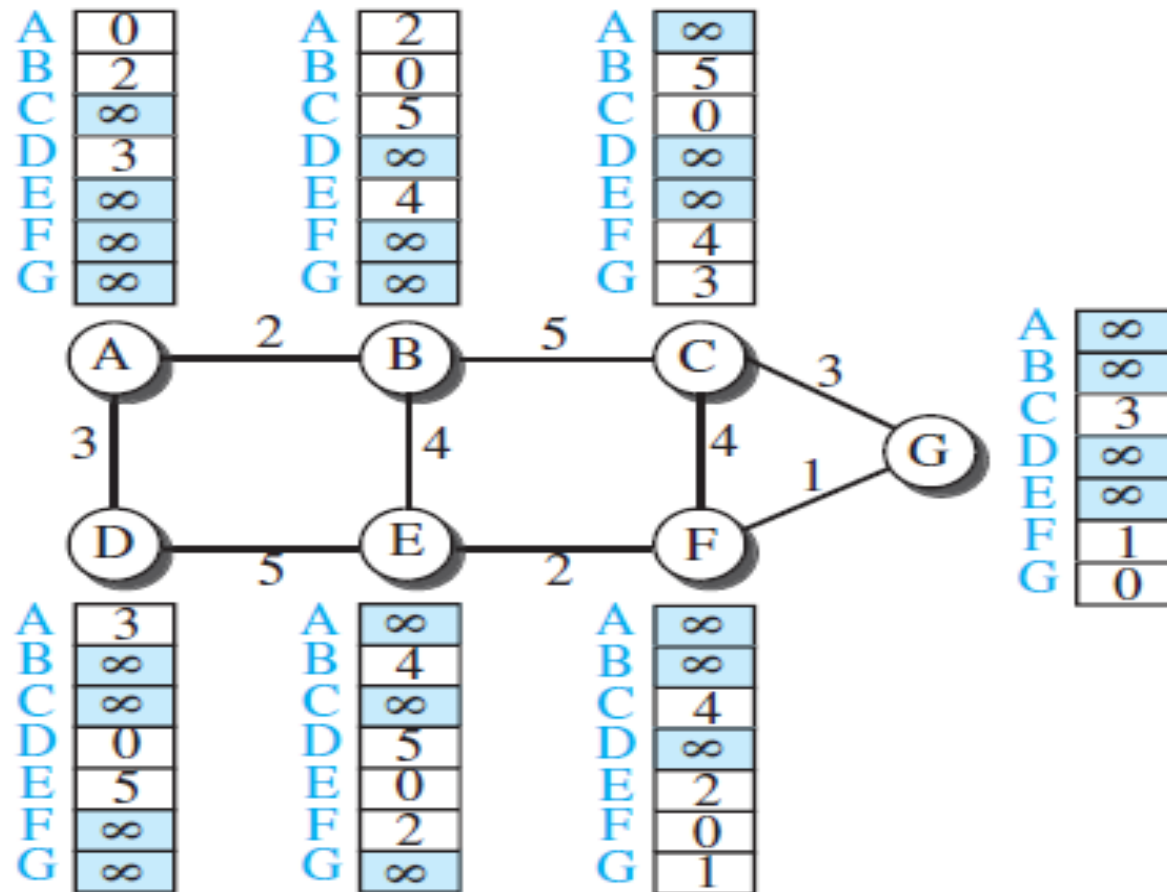
b. Distance vector for node A

Do these distance vectors represent least-cost paths?

They do, considering the limited information a node has.

When we know only one distance between two nodes, it is the least cost.

Initial Distance Vector for an internet in the Figure



Updating Distance Vectors

New B	Old B	A
A 2	A 2	A 0
B 0	B 0	B 2
C 5	C 5	C ∞
D 5	D ∞	D 3
E 4	E 4	E ∞
F ∞	F ∞	F ∞
G ∞	G ∞	G ∞

$B[] = \min(B[], 2 + A[])$

a. First event: B receives a copy of A's vector.

New B	Old B	E
A 2	A 2	A ∞
B 0	B 0	B 4
C 5	C 5	C ∞
D 5	D 5	D 5
E 4	E 4	E 0
F 6	F ∞	F 2
G ∞	G ∞	G ∞

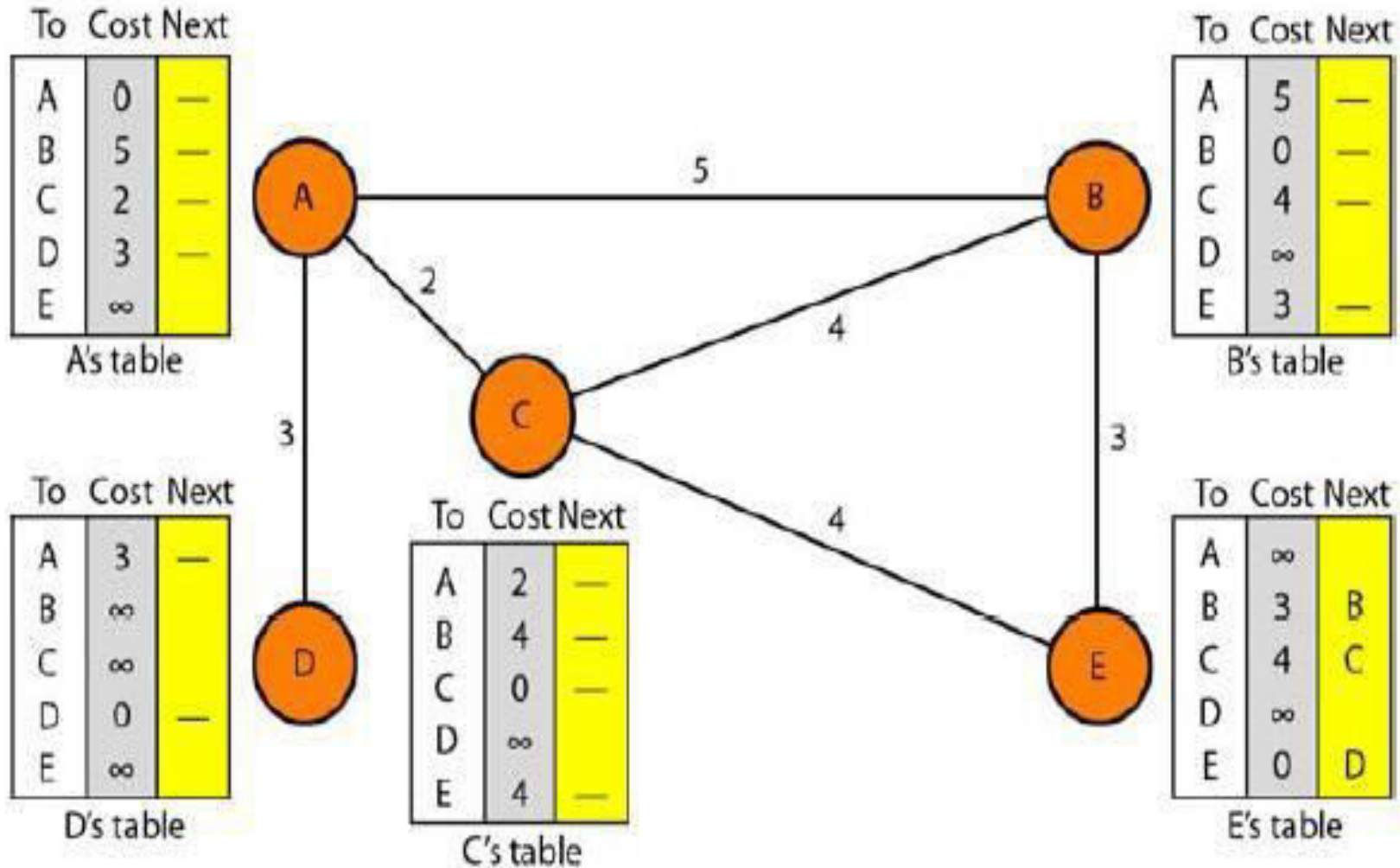
$B[] = \min(B[], 4 + E[])$

b. Second event: B receives a copy of E's vector.

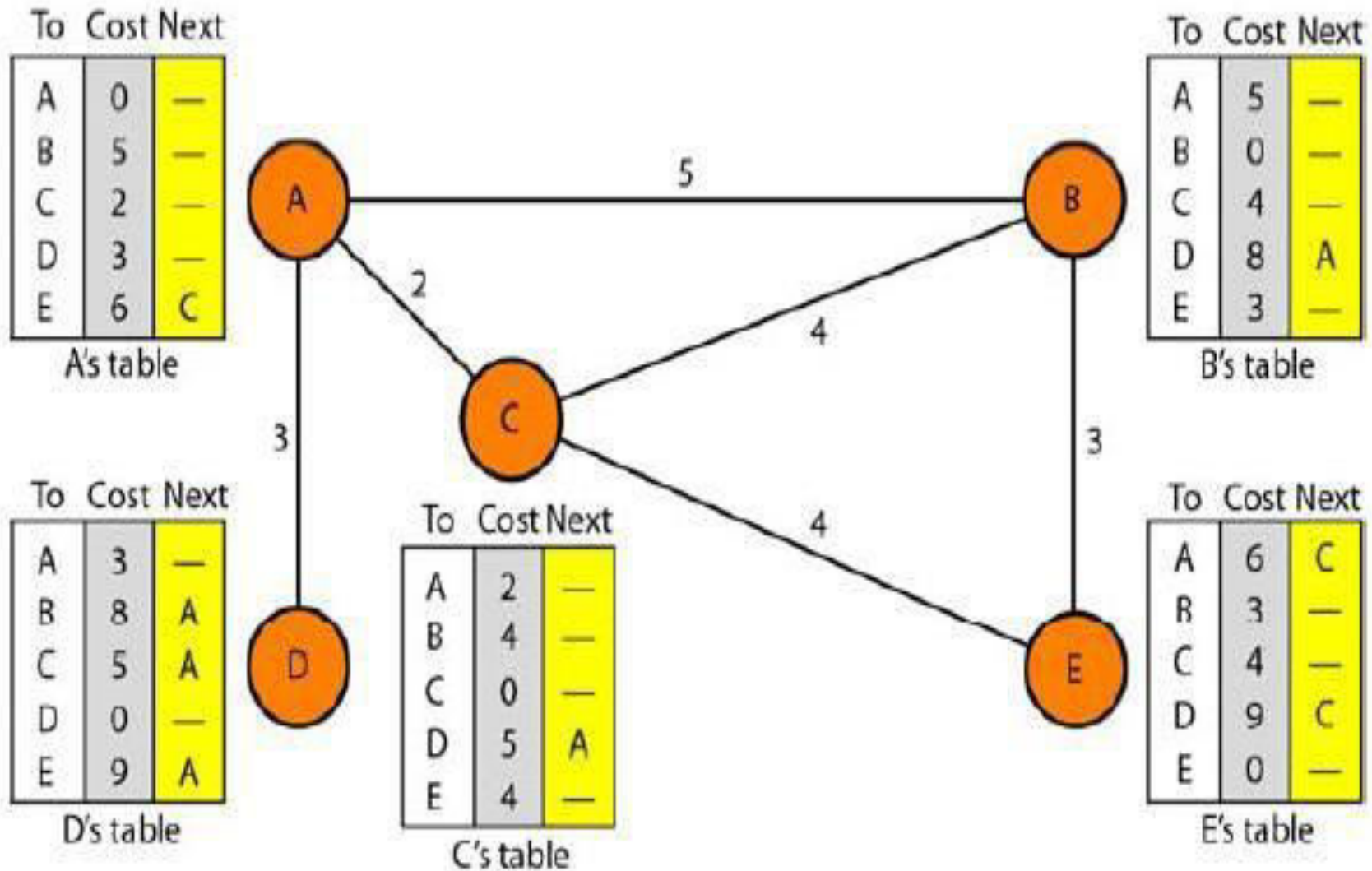
Note:

$X[]$: the whole vector

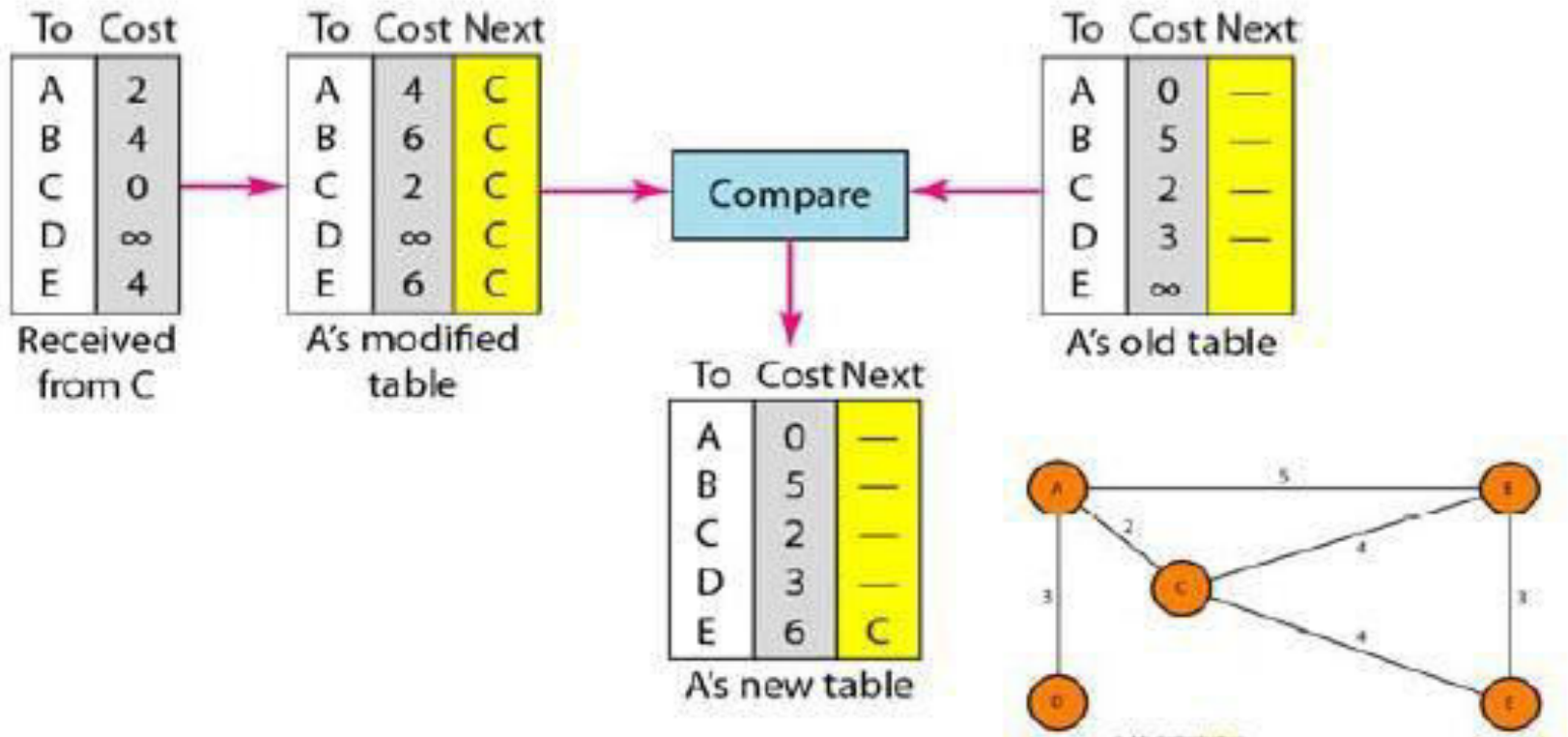
Distance Vector Routing: Table Initialization



Distance Vector Routing: Sharing



Distance Vector Routing: Updating



In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change

Updating Routing Table

- **If the next-node entry is different**
 - The receiving node chooses the row with the smaller cost
 - If there is a tie, the old one is kept
- **If the next-node entry is the same**
 - i.e. the sender of the new row is the provider of the old entry
 - The receiving node chooses the new row, even though the new value is infinity.

When to Share

➤ Periodic Update

- A node sends its routing table, normally 30 seconds, in a periodic update

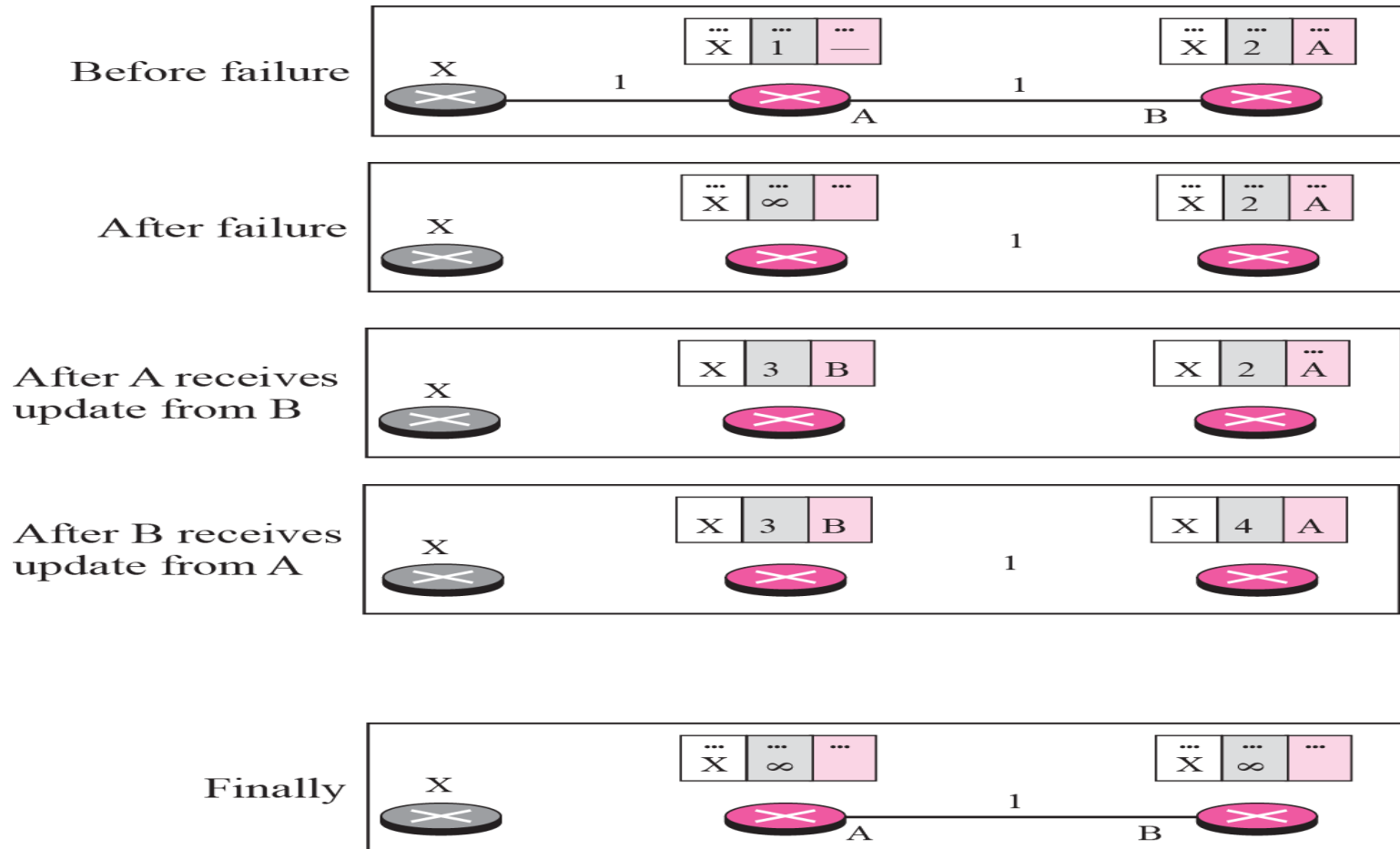
➤ Triggered Update

- A node sends its routing table to its neighbors any time when there is a change in its routing table
 - 1. After updating its routing table, or
 - 2. Detects some failure in the neighboring links

Count to Infinity

- A problem with distance-vector routing is that
 - any decrease in cost (good news) propagates quickly,
 - but any increase in cost (bad news) will propagate slowly.
- For a routing protocol to work properly, if a link is broken (cost becomes infinity),
 - every other router should be aware of it immediately,
 - but in distance-vector routing, this takes some time.
- **The problem is referred to as *count to infinity***
- It sometimes takes several updates before the cost for a broken link is recorded as infinity by all routers.

Count to Infinity: Two node loop



Most implementations define 16 as infinity

Solutions for Two-Node Instability

Split Horizon

- Split horizon is a method of preventing a routing loop in a network.
- **Simple principle:** Information about the routing for a particular packet is never sent back in the direction from which it was received.
- So, instead of flooding the table through each interface, each node sends only part of its table through each interface

